

LA HW

Included exercises: 1-2, 9, 12, 14-15, 37, 44

Exercises 1-8

In Exercises 1–8, compute the determinant of each matrix.

$$\begin{bmatrix} 6 & 2 \\ -3 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Original Exercise 1

$$\begin{bmatrix} 6 & 2 \\ -3 & -1 \end{bmatrix}$$

Original Exercise 2

$$\begin{bmatrix} 4 & 5 \\ 3 & -7 \end{bmatrix}$$

Original Exercise 9

the $(1, 2)$ -cofactor

Original Exercise 12

the $(3, 3)$ -cofactor

Exercises 13-20

In Exercises 13–20, compute the determinant of each matrix A by a cofactor expansion along the indicated row.

$$\begin{bmatrix} 2 & -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 2 \end{bmatrix}$$

Original Exercise 14

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$

along the second row

$$\begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Original Exercise 15

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$

along the third row

$$\begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Exercises 37-44

In Exercises 37–44, find each value of c for which the matrix is not invertible.

$$\begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$$

Original Exercise 37

Let V be a vector space over $\mathbb{R}$. Suppose $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8, v_9, v_{10}\}$ is a basis for V . Let T be a linear transformation from V to V defined by

$$T \begin{bmatrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 & v_8 & v_9 & v_{10} \end{bmatrix} = \begin{bmatrix} 3v_1 & 6v_2 & 9v_3 & 12v_4 & 15v_5 & 18v_6 & 21v_7 & 24v_8 & 27v_9 & 30v_{10} \end{bmatrix}.$$

Original Exercise 44

$$\begin{bmatrix} c & 9 \\ 4 & c \end{bmatrix}$$